Aishik Koley, X-B, Roll-17.

# Delhi Public School, Howrah

**PRE-BOARD II EXAMINATION (2025-2026)**

**Class-X**

*Care must be taken not to write anything on the question paper. All the questions must be attempted in the correct sequence.*

**Subject: - Science (Code No. 086)**

**Time:-3 Hours** **F.M.-80**

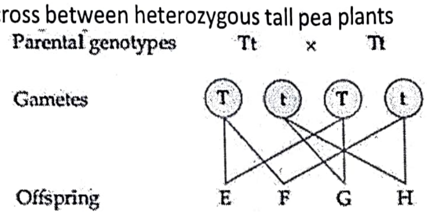
**General Instructions:**

i. **This question paper consists of 39 questions in 3 sections. Section A is Biology, Section B is Chemistry and Section C is Physics.**

ii. **All questions are compulsory. However, an internal choice is provided in some questions. A student is expected to attempt only one of these questions.**

## SECTION - A

**1.** The diagram shows a cross between heterozygous tall pea plants **1**




Which of the following statement is not correct?

A. Offspring E and H are both homozygous.

B. Offspring F and G are both heterozygous.

C. The phenotypes of offspring E, F and G are the same.

D. The ratio of different phenotypes in the offspring is 1 : 1.

**2.** The steps of the process of regeneration are given in a random order. Which of the following is the correct sequence of steps? **1**

i. Regeneration is carried out by specialized cells.

ii. From the mass of cells different cells undergo changes to become various cell types and tissues.

iii. The specialized cells proliferate and make large number of cells.

iv. The changes take place in an organised sequence referred to as development.

A. (i), (ii), (iii), (iv).

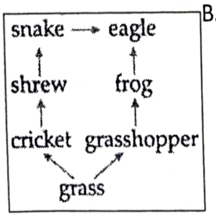
B. (i), (iv), (ii), (iii).

C. (iv), (ii), (iii), (i).

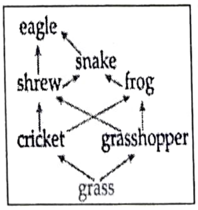
D. (i), (iii), (ii), (iv).

**3.** Which of these represents the correct food web of the grassland ecosystem? **1**

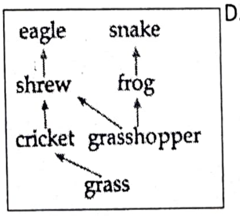
A.



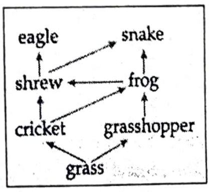

B.



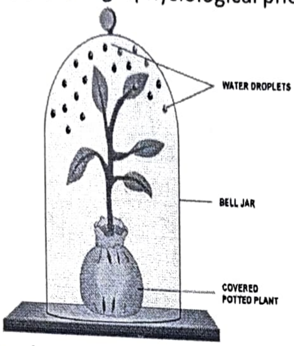

C.




D.



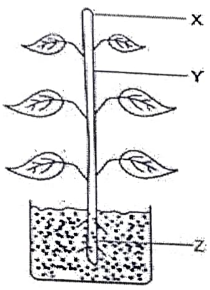
4. Observe the diagram given below. It is showing a physiological phenomenon in plants.




If the experiment is repeated using a **defoliated plant**, what will be the most likely observation?
A. Water droplets will still form on the bell jar
B. More water droplets will form
C. No water droplets will form
D. The bell jar will crack due to pressure

5. Which situation best explains Mendel's Law of Independent Assortment? 1
A. Crossing tall and dwarf plants
B. Inheritance of blood groups
C. Inheritance of seed colour and seed shape
D. Cloning of plants

6. Shown in the figure below is a plant in which ~~oxygen~~ Auxyn is synthesized at part X of the plant. Geeta took the potted plant and cut off the part X. Then she took the plant and kept it near a window with sunlight and observed it after 7 days. After 7 days she will observe that 1




A. Part Y grew and bent towards the window.
B. Part Z started growing upwards and out of the soil.
C. Part Y's growth significantly slows or stops.
D. Part Y grew upwards.

7. Study the table given below and identify the option with incorrect information. 1

| | Characters | Blood | Lymph |
|---|---|---|---|
| A | Cells present | RBC, WBC, and blood platelets | Blood platelets and WBC |
| B | Colour | Reddish in colour | Colourless |
| C | Location | Flows through blood capillaries | Flows through lymphatic capillaries |
| D | Function | Transports nutrients and oxygen from one organ to another | Transports nutrients from the tissue cells to the blood |

The following two questions consist of two statements – Assertion (A) and Reason (R). Answer these questions by selecting the appropriate option given below:

A. Both A and R are true, but R is not the correct explanation of A.
B. Both A and R are true, and R is the correct explanation of A.
C. A is true but R is false.
D. A is false but R is true.

8. **Assertion (A):** Plant hormones are produced at specific sites and act only at the same place. **1**
**Reason (R):** Plant hormones can be transported to other parts of the plant. (D)

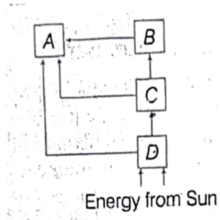
9. **Assertion (A):** Mendel's Law of Dominance explains why all $F_1$ plants of a monohybrid cross show only one parental trait. **1**
**Reason (R):** One of the alleles masks the expression of the other allele in a heterozygous condition. (B)

10. Mustard was growing in two fields P and Q. While field P produced brown coloured seeds, field Q produced yellow-coloured seeds. **2**
It was observed that in field P the offsprings showed only the parental trait for consecutive generations, whereas in field Q majority of the offsprings showed a variation in the progeny. What are the probable reasons for these?

11. <u>*Students to attempt either option A or B.*</u> **2**
A. How does ozone form and why is it important?

**OR**

B. We already know that food chain contains different organisms at different trophic levels in a typical ecosystem.
In the diagram of energy flow in an ecosystem given below, identify the secondary consumer and give reason for your choice.




12. In the following food chain only 2J of energy was available to the Peacocks. How much energy would have been present in grass? Justify your answer. **2**
Grass → Grasshopper → Frog → Snake → Peacock
10000 2000 200 20

13. A. Why do doctors often measure TSH levels alongside thyroxine levels to diagnose thyroid disorders? **3**
B. State the consequences of low TSH level?
C. Why does a deficiency of thyroid hormone in childhood lead to mental retardation, while the same deficiency in adults mainly affects metabolism? (RRyy)

14. If two pea plants having round and green seeds (RRyy) are crossed, identify the percentage of the following with respect to the $F_1$ generation: **3**
A. Gametes having both the round and yellow seed traits.
B. Offspring having the same genotype as the parents.
C. Offspring having the same phenotype as the parents.

15. **Read the following case and answer the following questions:**
To sustain life Our body must produce enough energy which is produced by burning of food molecules in the presence of oxygen. Oxidation of food molecules produces carbon dioxide and water. To take in oxygen and to expel out carbon dioxide, respiratory system is present in human beings. The respiratory system starts at the nose and mouth and continues through the Airways and the lungs. The energy released during the process of respiration is used to make an atp molecule from atp and inorganic phosphate.
A. Name and draw the balloon like structures present in the lungs.
B. Write the pathway for the breakdown of glucose in yeast cell and in mitochondria
*Attempt either subpart C or D.*
C. Fat gives us more energy than glucose, then why is glucose considered as the preferred respiratory fuel in humans?

OR

D. A person has a blocked nasal passage and starts breathing through the mouth. How will this affect respiration in the long run?

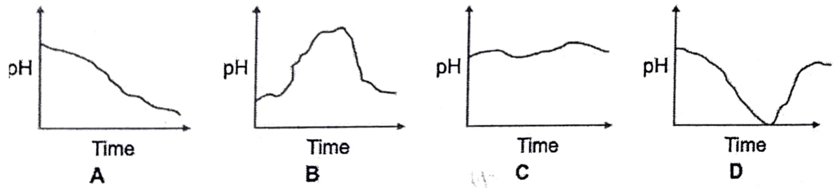
16. *Attempt either option A or B.* 3+2
A. I.Given below are certain situations. Analyse and describe its possible impact on a person.
   i. The menstrual cycle in females ceases during pregnancy.
   ii. Vas deferens of a man is plugged.
   iii. Prostate and seminal vesicles are not functional.
II. Draw a diagram to show germination of pollen on stigma.

OR

B. I. List the advantages of growing grapes or banana plants through vegetative propagation.
II. A potato is cut into a number of small pieces. These pieces are then placed only on wet cotton kept in a tray. After few days some of the potato pieces give rise to fresh green shoots and roots. Why?
III. Draw the various stages of binary fission in *Leishmania*

## SECTION - B

17. Which of these graphs shows how the pH of milk changes as it forms curd? 1




18. During a chemistry experiment, Rohan mixed an aqueous solutions of sodium chloride and silver nitrate inside a test tube. A white precipitate was formed. Which of the following is the correct balanced chemical equation for this reaction? 1
A. $NaCl + AgNO_3 \rightarrow NaNO_3 + AgCl$
B. $2NaCl + AgNO_3 \rightarrow Na_2NO_3 + AgCl_2$
C. $NaCl_2 + 2AgNO_3 \rightarrow Na(NO_3)_2 + 2AgCl$
D. $NaCl + Ag_2NO_3 \rightarrow NaAg + AgCl_2$

19. The yellow colour of turmeric changes to red on addition of soap solution. When substance P is added to turmeric, there is no change in colour. Which of the following is true about the substance P? 1
A. P is an acid.
B. P is not a salt.
C. P is not a base.
D. P is a neutral substance.

1

Listed here is the reactivity of certain metals:

| Metal | Reaction with air | Reaction with water | Reaction with dilute acids |
|---|---|---|---|
| Gold | Does not oxidize or burn | No reaction | No reaction |
| Sodium | Burns vigorously to form oxide | Violent reaction | Violent reaction |
| Zinc | Burns to form oxides | Reacts on heating with water | Reacts to produce hydrogen |
| Platinum | Does not oxidize or burn | No reaction | No reaction |

Which of the above metals are likely to be obtained in their pure states from the Earth's crust?

A. Gold only
B. Sodium only
C. Gold and platinum
D. Zinc and sodium

21. A yellow powder X gives a pungent smell if left open in air. It is prepared by the reaction of dry compound Y with chlorine gas. X is used for disinfecting drinking water. Identify X and Y. 1

A. X=$Ca(OH)_2$, Y=$CaOCl_2$
B. X= $CaOCl_2$, Y= $Ca(OH)_2$
C. X= $CaOCl_2$, Y= NaOH
D. X=NaOH, Y=$CaOCl_2$

22. A piece of zinc (Zn), a reactive metal was dropped into a test tube containing a substance. A zinc salt was formed and hydrogen gas was liberated. This is shown in the equation below. 1

Zn + _____→ zinc salt + $H_2$ gas

Which of the following can be the substance that zinc was dropped into?

(i) Water
(ii) Hydrochloric acid
(iii) A solution of a zinc salt

A. Only (i)
B. Only (ii)
C. Only (iii)
D. Either (i) or (iii)

23. Which of the following compounds contain maximum number of water of crystallization in its crystal form in one molecule? 1

A. $FeSO_4$
B. $CuSO_4$
C. $CaSO_4$
D. $Na_2CO_3$

**The following question consists of two statements – Assertion (A) and Reason (R). Answer these questions by selecting the appropriate option given below:**

**A. Both A and R are true, and R is the correct explanation of A.**
**B. Both A and R are true, and R is not the correct explanation of A.**
**C. A is true, but R is false.**
**D. A is false, but R is true.**

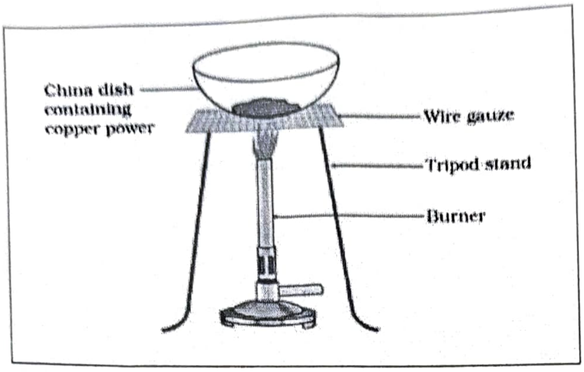
24. **Assertion (A):** In a homologous series of alcohols, the formula for the second member is $C_2H_5OH$, and the formula for the third member is $C_3H_7OH$. 1
**Reason (R):** The difference between the molecular masses of the two consecutive members of a homologous series is 16u.

25. A teacher asks her students to identify a metal, M. She gives them the following clues to help. 2

- Its oxide reacts with both HCl and NaOH.
- It does not react with hot or cold water but reacts with steam.
- It can be extracted by electrolysis of its ore.

A. Identify the metal.
B. What will happen if the metal reacts with iron oxide?

26.




A. Observe the given figure and write what happens when copper powder is heated in a china dish. Explain your observation.

B. What will happen if hydrogen gas is passed over the heated material obtained in the above observation? Identify the substance oxidised and the substance reduced in the reaction of this case.

27. *Attempt either option A or B.* 3

A. Maya had a compound of carbon and hydrogen whose formula was $C_3H_4$.

I. What type of flame will this compound give on combustion?

II. What will be the IUPAC name and structural formula of the compound?

III. Maya took few drops of reddish solution of bromine water in a test tube and shook it with the hydrocarbon that she had with her. What was her observation?

**OR**

B. I. Name the products formed when ethanoic acid reacts with sodium carbonate. Write the balanced chemical equation.

II. Write the IUPAC name and structural formula of the first member of the ketone series.

28. Study the following clues and answer the following questions by analyzing them. 4

- Substance 'C' is commonly used in whitewashing of walls. $Ca(OH)_2$
- 'C' is formed when substance 'B' reacts with water. $CaO$
- Substance 'B' is obtained by heating substance 'A' at a high temperature in a lime kiln. $CaCO_3$
- Substance 'A' is a naturally occurring compound found in limestone, marble, and chalk. $CaCO_3$
- Substance 'C' reacts with carbon dioxide to form substance 'A' again. $CaCO_3$

A. Identify the CORRECT statement regarding substance 'B'.

(a) It reacts slowly with water

(b) It is used for fast cooking

(c) It reacts vigorously with water producing heat

(d) It is obtained from acids

*Attempt either part B or C.*

B. Write the sequence of chemical reactions connecting 'A', 'B' and 'C'.

**OR**

C. Identify substances 'A', 'B', and 'C' and also identify the type of reaction when 'B' is formed from 'A'.

D. The reaction of substance 'B' with water to form 'C' is an example of:

(a) Decomposition reaction

(b) Combination reaction

(c) Displacement reaction

(d) Neutralization reaction

29. *Attempt either option A or B.* 5

A. I. An unsaturated organic compound 'P' with molecular mass 28u, on addition with 1 mole of hydrogen in presence of nickel, changes to a saturated hydrocarbon 'Q'.

i. Write down the balanced chemical equation for the reaction involved.

ii. Write the IUPAC name and structural formula of the fifth member of the homologous series to which compound 'Q' belongs.

iii. What happens when compound 'Q' undergoes complete combustion?

II. Compound 'A' works well with hard water. It is used for making shampoos and products for cleaning clothes. 'A' is not 100% biodegradable and causes water pollution. Compound 'B' does not work well with hard water. It is 100% biodegradable and does not create water pollution. Identify A & B.

**OR**

B. I. Give a chemical test to distinguish between:

(i) Ethane and ethene

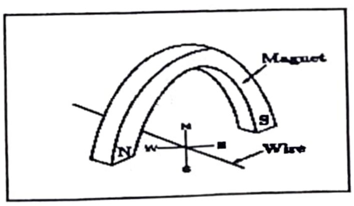
(ii) Ethanol and ethanoic acid

(iii) Soaps and Detergents

II. An organic compound X with a molecular formula $C_2H_6O$ undergoes oxidation in presence of alkaline $KMnO_4$ to form a compound Y. X on heating in presence of conc. Sulphuric acid gives Z with a general formula $C_nH_{2n}$. Y can also react with X in presence of conc. $H_2SO_4$ to form another substance R with fruity smell. Identify X, Y, Z and R.

## SECTION - C

**30.** A current carrying copper wire is held between the poles of a magnet. The current in the wire can be reversed. The pole of the magnet can also be changed over. In how many of the four directions shown in the figure can the force act on the wire? 1




A. 1 B. 2 C. 3 D. 4

**31.** Four students A,B,C and D did their experiment of finding the focal length of convex lens by obtaining image of a distant object as follows: 1

- Student A used the window grill in the laboratory as the object and a white paper sheet held in hand as the screen.
- Student B used a distant tree in the shade and a white thick board held in a stand as the screen.
- Student C used a well illuminated laboratory window grill as the object and a white paper sheet held in a stand as the screen.
- Student D used a well illuminated distant tree as the object and a white thick board held in a stand as the screen. Which student has used the correct method for performing experiment out of the above setups?

A. Student A B. Student B C. Student C D. Student D

The following question consists of two statements – Assertion (A) and Reason (R). Answer these questions by selecting the appropriate option given below:

A. Both A and R are true, and R is the correct explanation of A.

B. Both A and R are true but R is not the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

32. **Assertion (A):** On changing the direction of flow of current through a straight conductor, the direction of a magnetic field around the conductor is reversed. **1**

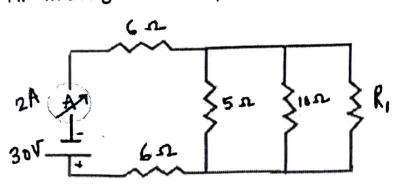
**Reason (R):** The direction of magnetic field around a conductor can be given in accordance with left hand thumb rule.

33. A convex lens of focal length 40 cm and a concave lens of focal length 50 cm are placed in contact with each other. Calculate: (i) the power of the combination, (ii) focal length of the combination. **2**

0.5D 2m

34. *Attempt either option A or B.* **2**

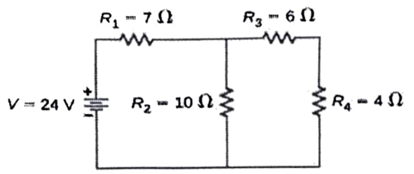
A. In the given circuit, if the current reading in the ammeter A is 2A, what would be the value of $R_1$?




**2**

OR

B.




Calculate the total resistance of the circuit and find the total current in the circuit.

35. Rohit wants to have an erect image of an object using a converging mirror of focal length 40 cm. **3**

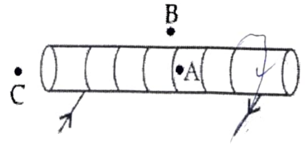
A. Specify the range of distance where the object can be placed in front of the mirror. Justify.

B. Draw a ray diagram to show image formation in this case.

C. State one use of the mirror based on the above kind of image formation.

36. For the current carrying solenoid as shown- **2+1**

A. Draw magnetic field lines and give reason to explain that out of the three points A, B and C, at which point the field strength is maximum and at which point it is minimum?




B. List two ways of increasing the strength of an electromagnet if the material of the electromagnet is fixed.

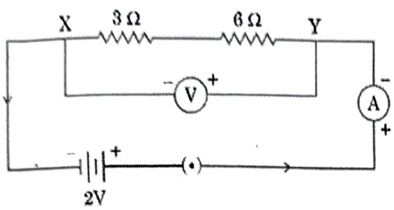
37. A. Why is an alternating current (A.C.) considered to be advantageous over direct current (D.C.) for the long distance transmission of electric power? 3

B. How is the type of current used in household supply different from the one given by a battery of dry cells?

C. How does an electric fuse prevent the electric circuit and the appliances from a possible damage due to short circuiting or overloading?

38. *Attempt either subpart C or D.* 1+2

Study the circuit shown in which two resistors X and Y of resistances 3 Ω and 6 Ω respectively are joined in series with a battery of 2V. +1




A. Draw a circuit diagram showing the above two resistors X and Y joined in parallel with same battery and same ammeter and voltmeter.

B. In which combination of resistors will the (i) Potential difference across X and Y and (ii) current through X and Y, be the same? Justify your answer.

C. Find the current drawn from the battery by the series combination of the two resistors X and Y.

**OR**

D. Determine the equivalent resistance of the parallel combination of the two resistors X and Y.

39. *Attempt either option A or B.* (1+1

A. (I) The power of a lens 'X' is – 2·5 D. Name the lens and determine its focal length in cm. For which eye defect of vision will an optician prescribe this type of lens as a corrective lens? +2 +1)

(II) "The value of magnification 'm' of a lens is – 2." Using new Cartesian Sign Convention and considering that an object is placed at a distance of 20 cm from the optical centre of this lens, state : (a) the nature of the image formed; size of the image compared to the size of the object, (b) position of the image .

(III) The numerical values of the focal lengths of two lenses A and B are 10 cm and 20 cm respectively. hich one of the two will show higher degree of convergence/divergence? Give reason to justify your ıswer.

**OR**

B. (I) Draw and label a ray diagram to show the refraction of a ray of light through a rectangular glass slab when it falls obliquely from air into glass.

(II) State Snell's law of refraction of light.

(III) Differentiate between the virtual images formed by a convex lens and a concave lens on the basis of : (a) object distance, and (b) magnification. (2+1 +2)